

C3: Study Notes Discussion 3: Custom Types and Arrays

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Date: 27 August 2026

Overview

Extend your study notes with the concepts and tools from the Structuring Data and Working with Multiples chapters of our programmer's field guide, as indicated in the checklist at the end of this document.

Do the following:

- If you can, print the checklist from the end of this document and paste it into the next empty pages of your workbook. As before, use this to tick off concepts covered as you go.
- Update the checklist below as you progress.
- Move on to Reflect when you have completed all of the concepts from the checklist.

Make it useful!

Remember this is your chance to develop and communicate your understanding. Use this to help develop and discuss your understanding with your peers and the teaching team. Interacting with others and discussing these ideas will help you develop your own

Reflect:

- What is the most important thing you learned from this and why?

> I learned many thing in this, its very important to improve our programming skill with using array (which std::vector in c++), struct and enum. This can restructure our code to be clean, without repetition, and also work together with function, procedure, while loop and for loop. We can't write many project without this.

- What did you find the most interesting or challenging about arrays?

> I found the array is the interesting and challenging one. Because it is very useful to grouping a data. Otherwise, it related to algorithm, many algorithm using array and working with for loop to operation it.

- Why did you select the different pages you choose to photograph and include in your portfolio?

> First page I choose is about the mind map that combine all the concept, I like it because it shown the connectivity about the programming. Second one is about the array, I choose it because it is very interesting, array can be easy or very depend on what

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I did not use any AI for this task

Submit:

- Export this worksheet as a PDF.
- Take a photograph of two pages from your study notes: one that is representative of your work in general, and one that you are particularly proud of or feel was particularly influential in your learning.
- Upload your submission to OnTrack.
- Engage with feedback and discussion to get this signed off:

Checklist:

- Mind map of **structured data** connecting the following terms (1 page):
 - Data type
 - Struct
 - Enum
- Exploration of **struct** construct showing: (1 page)
 - Visualisation, metaphor, or illustration showing how you “see” these
 - Connect with related concepts:
 - Fields
 - Data Type
 - Variable
 - Expression
 - Syntax for struct declaration / creation
 - Annotated example struct declaration
 - Annotated example variable declaration, with your struct’s type
 - Illustration of how struct values are organised in memory
 - Notes on when and why to use structs
- Exploration of **enum** construct showing: (1 page)
 - Visualisation, metaphor, or illustration showing how you “see” these
 - Connect with related concepts:
 - Values
 - Data Type
 - Variable
 - Expression
 - Syntax for enum declaration / creation
 - Annotated example enum declaration
 - Annotated example variable declaration, with your enum’s type
 - Illustration of how struct values are organised in memory
 - Notes on when and why to use structs

- Exploration of **array** construct showing: (1 page)
 - Visualisation, metaphor, or illustration showing how you “see” these
 - Connect with related concepts:
 - Data Type
 - Array
 - Index
 - Expression
 - `dynamic_array`
 - `fixed_array`
 - Pass by reference
 - Syntax for array declaration / creation
 - Annotated example array declaration
 - Annotated example array access
 - Illustration of how array values are organised in memory
 - Notes on when and why to use arrays (`dynamic_array` and `fixed_array`)
- Mind map of **iteration over arrays** connecting the following terms (1 page):
 - Array
 - Index
 - For loop
- Exploration of **array** of **structs** usage showing: (1 page)
 - Annotated example array declaration
 - Annotated example array access
 - Illustration of how array values are organised in memory
- Exploration of **array** usage: (1 page)
 - Illustrate connection between actions with the computer and values in the array